

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B.Tech

Branch : Computer Engineering/Computer Science & Engineering

Subject Code & Name: BTCOE703C

Big Data Analytics

Semester : VII

Max Marks: 60

Date: 10/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Transaction of data of the bank is a type of -----.	L1/CO1	1
	a.Unstructure d data b.Structured data c. Social data d. Machine Data		
2	_____ is the slave/worker node and holds the user data in the form of Data Blocks.	L1/CO3	1
	a.DataNode b. NameNode c.Data block d. Replication		
3	The _____ is responsible for allocating resources to the various running applications subject to familiar constraints of capacities, queues.	L2/CO2	1
	a.Manager b. Master c. Tracker d. Scheduler		
4	How much will bytes for 1Petabytes	L3/CO1	1
	a.1024bytes b.1024 ⁵ bytes c.1024 ⁴ bytes d. 1024 ⁶ bytes		
5	In case of Yet Another Resource Negotiator (YARN), the _____ User Interface provides host and port information.	L2/CO2	1
	a.Data Node b. Name Node c. Replication d. Resource Manager		
6	Kafka is comparable to traditional messaging systems such as _____	L2/CO3	1
	a.Impala b. Zookeeper c. ActiveMQ d.BigTop		
7	Mapper implementations are passed the JobConf for the job via the _____ method.	L3/CO2	1
	a. JobConfigure .configure b. JobConfigurable. configurable c. JobConfigurabl e.configure d. JobMapConfig ure.configure		
8	Mahout provides an implementation of a _____ identification	L2/CO4	1

	algorithm which scores collocations using log-likelihood ratio.					
	a.collocation	b.compaction	c.collection	d. Corelation		
9	What is mean by “collection” in MongoDB?				L2/CO5	1
	a.A group of tables	b. A group of documents	c. A single document	d. A type of index		
10	_____ leverages Spark Core fast scheduling capability to perform streaming analytics.				L2/CO3	1
	a.MLlib	b. GraphX	c. Index	d. Spark Streaming		
11	Which of the following format is supported by MongoDB?				L2/CO5	1
	a. BSON	b. XML	c.SQL	d. NoSQL		
12	_____ is a distributed graph processing framework on top of Spark.				L2/CO4	1
	a. Spark	b. ML library	c.GraphX	d. BSON		
Q. 2	Solve the following.					12
A)	What is a big data? Discuss the different sources of big data.				L2/CO1	6
B)	Describe the various characteristics of Big data.				L2/CO1	6
Q.3	Solve the following.					12
A)	With neat Sketch explain HDFS architecture in detail.				L2/CO2	6
B)	Explain the Map Reduce programming model. Solve the example of Word Count for 1.This is an Apple. 2.An Apple is red in colour.				L3/CO2	6
Q. 4	Solve Any Two of the following.					12
A)	Why is Real-time Big Data Pipeline So Important Nowadays?				L2/CO3	6
B)	Explain the Role and working of APACHE KAFKA.				L2/CO3	6
C)	Discuss the typical use cases of streaming data.				L2/CO3	6
Q.5	Solve Any Two of the following.					12
A)	What is Machine Learning? Why the Machine Learning is important in Big Data processing?				L2/CO4	6
B)	What is Graph processing? Discuss the Graph processing algorithms in big data.				L2/CO4	6
C)	Explain the K-Means clustering Algorithm with example.				L2/CO4	6
Q. 6	Solve Any Two of the following.					12
A)	Discuss the advantages of using Java script shell for MongoDB queries				L3/CO5	6

B)	Explain the CRUD operations in MongoDB with example.	L2/CO5	6
C)	What is the NoSQL database? Explain different types of NoSQL databases.	L2/CO5	6
*** End ***			

51647368

51647368

51647368

51647368

51647368

51647368

51647368

51647368

51647368